

Predicting Stock Downside Risk

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Introduction

The stock market is a public marketplace where people buy and sell shares, small pieces of ownership in companies like Apple or Amazon. While investors aim to grow wealth through company success, market prices are unpredictable and fluctuate constantly. Most people focus on guessing if a stock will go up or down, but for everyday investors, the greatest challenge is not knowing the magnitude of a potential loss. While small changes are normal, sudden and large drops can happen quickly. By developing a predictive model that identifies the specific percentage a stock might decline over a five day window, we can provide a warning system. This allows individuals to assess their risk exposure and protect themselves before significant losses occur.

Objectives

Our overall goal of this project was to determine to what extent historical market indicators and company financial fundamentals can be used to predict the magnitude of a stock's downside risk, measured as the maximum percentage decline over the next 5 trading days.

- **Build a Dataset:** Collect and Process time series data from the Alpha Vantage API for high-volume technology companies.
- **Calculate Market Vital Signs:** Use technical indicators, such as RSI and Bollinger Bands, to measure price momentum and market instability.
- **Develop and Train Models:** Use machine learning to predict and identify the relationship between market variables and the magnitude of a stock's next 5-day decline.
- **Analyze Feature Significance:** Determine which indicators are the most reliable predictors of a drop.
- **Evaluate Model Performance:** Compare the model's predictions against tests to validate its accuracy.

Methods/Collected Data

Linear Regression (Best Performing Model)

- Simple and easy to interpret baseline model
- Predicts downside risk using a weighted combination of features
- Allows us to clearly see how each feature impacts predictions
- Performed the best, suggesting the relationship in the data is mostly linear

Random Forest

- Uses multiple decision trees and averages their predictions
- Captures more complex patterns and interactions between features
- Reduces overfitting compared to a single decision tree
- Provides feature importance to identify key predictors
- Did not outperform Linear Regression in this project

Gradient Boosting

- Builds models sequentially, correcting previous errors step-by-step
- Strong at capturing complex relationships in structured data
- Typically more powerful than simpler models
- Performed reasonably well but worse than Linear Regression, indicating the data does not require high complexity

Collected Data Summary

- Dataset includes 380 days of stock data from major tech companies
- Features include price, volume, and technical indicators (RSI, Bollinger Bands)

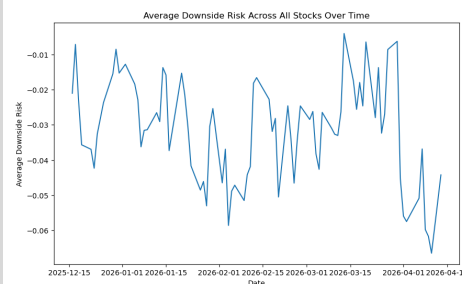
Key Insights:

- Downside risk changes over time, with some periods showing larger potential losses
- Most losses fall between -2% and -6%, with a few extreme drops
- Price-related features are strongly connected, while indicators like RSI add extra signals
- Overall, the data shows clear patterns that can help predict short-term risk

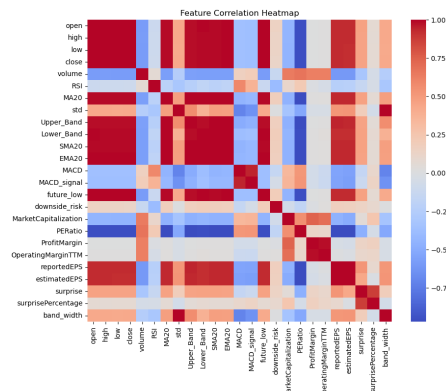
Results

The evaluation of our predictive models demonstrated that integrating technical and fundamental features significantly improves market risk forecasting.

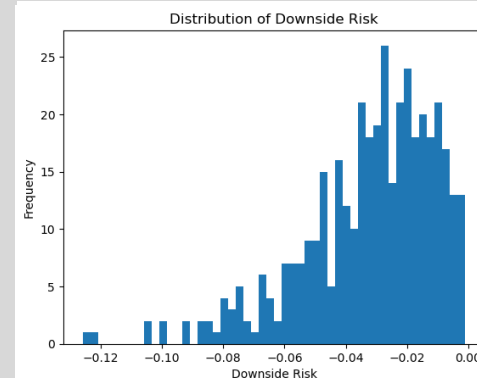
- **Best Performing Model:** The Linear Regression model utilizing the "Full" feature set achieved the highest accuracy with an R^2 of 0.526 in time-based testing.
- **Prediction Power:** The results suggest that our engineered features explained 52.6% of the variance in downside price movements.
- **Low Error Margin:** The model's predictions were usually within 1.4% of the actual drop, indicating a strong alignment between predictions and actual market outcomes.
- **Feature Impact:** Models using "Basic" features (price/volume) failed to provide predictive value. This suggests that indicators like RSI (how fast people are buying/selling) and Bollinger Bands (how much the price is jumping around) are essentials for quantifying risk.
- **Insight:** By using a Time-Based Split, we demonstrated that our model is identifying consistent technical triggers rather than simply memorizing historical noise.



This visualization shows the average downside risk across 5 stocks we chose.



This visualization shows the Correlation Heatmap of all the features we used in this project.



This visualization show the downside risk distribution of the 5 selected stocks

Conclusions

Overall this model can help investors and market analysts better understand and manage risk in the stock market. Instead of predicting only gains, our model focuses on estimating how much a stock could potentially drop in the future. This model works by combining technical indicators such as RSI with historical stock price data. Ultimately our project also reinforces the importance of acknowledging risk awareness in financial markets and decision making. Incorporating predictive risk models allows investors to make more informed decisions. Further development to expand the dataset, including adding real-time data, and more advanced models would improve accuracy of our model and its real-world applicability.

References

Alpha Vantage. (2026). *Time Series Daily (TIME_SERIES_DAILY)* (Data set). Alpha Vantage. alphavantage.co